

1. Operations with functions

Let f and g be real-valued functions with domains D_f and D_g , and let $c \in \mathbb{R}$ be a constant. A function includes both a rule and its domain; every operation must specify where it is defined.

Addition, multiplication, scalar multiplication, and division

The first three operations below are defined by ordinary arithmetic on function values. In a quotient, the denominator must also be nonzero.

Operation	Rule	Domain
Addition	$(f + g)(x) = f(x) + g(x)$	$D_f \cap D_g$
Multiplication	$(fg)(x) = f(x)g(x)$	$D_f \cap D_g$
Scalar multiplication	$(cf)(x) = cf(x)$	D_f
Division	$(f/g)(x) = \frac{f(x)}{g(x)}$	$\{x \in D_f \cap D_g : g(x) \neq 0\}$

Here $D_f \cap D_g$ means the inputs belonging to both domains. Subtraction can be written $f - g = f + (-1)g$.

Example

If $f(x) = x^2 - 1$ and $g(x) = x - 1$, both on $\mathbb{R}$, then

$$(f/g)(x) = \frac{x^2 - 1}{x - 1} = x + 1 \quad (x \neq 1).$$

The quotient still has domain $\mathbb{R} \setminus \{1\}$. Simplifying the formula does not supply a value at an excluded input. Likewise, $0f$ is the zero function on D_f , not an automatic extension to all real inputs.

Composition

Define the composition $f \circ g$, read “ f composed with g ,” by

$$(f \circ g)(x) = f(g(x)), \quad D_{f \circ g} = \{x \in D_g : g(x) \in D_f\}.$$

First evaluate $g(x)$; then use its output as the input of f . The notation $f \circ g$ denotes composition, whereas fg denotes the product of the two values at the same input.

Example

Let $f(x) = \sqrt{x}$, with domain $[0, \infty)$, and $g(x) = x - 1$, with domain $\mathbb{R}$. Then

$$\begin{aligned} (f \circ g)(x) &= \sqrt{x - 1}, & D_{f \circ g} &= [1, \infty), \\ (g \circ f)(x) &= \sqrt{x} - 1, & D_{g \circ f} &= [0, \infty). \end{aligned}$$

The first needs $x - 1 \geq 0$; the second needs $x \geq 0$. In general, $f \circ g \neq g \circ f$.

The identity function. On a set D , define $\text{id}_D(x) = x$. An inverse function will undo a function by composition, producing the identity on the appropriate domain.

2. Inverse functions

An inverse reverses the input-output relationship of a function. It exists when each output corresponds to exactly one input.

Exponentials and squaring. Since 2^x is strictly increasing, $2^u = 2^v$ implies $u = v$. Squaring does not have this property on $\mathbb{R}$: $1^2 = (-1)^2$.

One-to-one. A function f with domain A is one-to-one if

$$f(x_1) = f(x_2) \implies x_1 = x_2 \quad (x_1, x_2 \in A).$$

Equivalently, every horizontal line meets its graph at most once. A strictly increasing or strictly decreasing function is one-to-one.

Definition of the inverse

If f is one-to-one and has range $B = f(A)$, its inverse is the function $f^{-1} : B \rightarrow A$ defined by

$$f^{-1}(y) = x \iff f(x) = y.$$

Consequently,

$$f^{-1}(f(x)) = x \quad (x \in A), \quad f(f^{-1}(y)) = y \quad (y \in B).$$

In composition notation, $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$. The inverse domain is the original range, and the inverse range is the original domain. The notation f^{-1} does not mean $1/f$.

Example

On $\mathbb{R}$, $f(x) = (x-1)^2 + 2$ is not one-to-one: $f(0) = f(2) = 3$. Restrict its domain to $[1, \infty)$. It is then increasing, with range $[2, \infty)$. Solve

$$y = (x-1)^2 + 2, \quad x \geq 1 \implies x = 1 + \sqrt{y-2}.$$

Thus $f^{-1}(y) = 1 + \sqrt{y-2}$, $y \geq 2$. The negative square-root branch would return inputs outside the chosen domain.

Check both compositions. For $x \geq 1$, $f^{-1}(f(x)) = 1 + \sqrt{(x-1)^2} = 1 + |x-1| = x$. For $y \geq 2$, $f(f^{-1}(y)) = (\sqrt{y-2})^2 + 2 = y$.

The other branch. If the same formula is instead restricted to $(-\infty, 1]$, its inverse is

$$f^{-1}(y) = 1 - \sqrt{y-2}, \quad y \geq 2,$$

with range $(-\infty, 1]$. The domain restriction determines the branch.

3. Logarithms are inverse exponentials

For $a > 0$, $a \neq 1$, the exponential $x \mapsto a^x$ is strictly monotone and has range $(0, \infty)$. These facts give the following definition.

Definition of logarithms

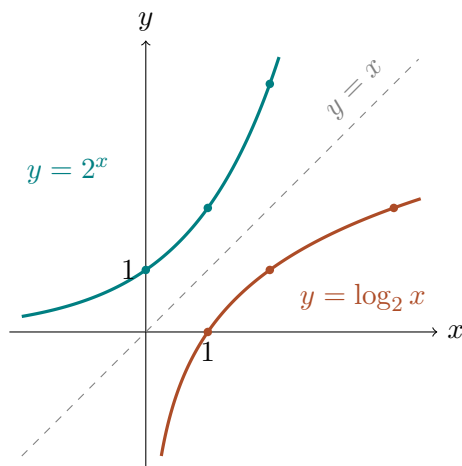
The logarithm $\log_a : (0, \infty) \rightarrow \mathbb{R}$ is the inverse of the exponential function $x \mapsto a^x$. Thus, for $u > 0$, $\log_a u$ is the unique real number x such that

$$\log_a u = x \iff a^x = u.$$

$$\log_a(a^x) = x \quad (x \in \mathbb{R}), \quad a^{\log_a u} = u \quad (u > 0).$$

The logarithm has domain $(0, \infty)$ and range $\mathbb{R}$.

For example, $\log_3 27 = 3$, $\log_4 2 = 1/2$, and $\log_{1/2} 8 = -3$. A logarithm may be negative; its argument must be positive. The base 1 is excluded because 1^x is constant.



Reflect in $y = x$.

The point (x, y) on a function becomes (y, x) on its inverse. Thus $(0, 1)$ becomes $(1, 0)$.

For $a > 1$, $\log_a$ is increasing; for $0 < a < 1$, it is decreasing.

For $a > 1$, values of $\log_a u$ become arbitrarily negative as positive u becomes arbitrarily small. The vertical axis is an asymptote.

The natural base. We may define

$$e = \sup \left\{ \left(1 + \frac{1}{n} \right)^n : n = 1, 2, 3, \dots \right\} \approx 2.718281828.$$

The set is nonempty and bounded above by 3, a fact we accept here. Write $\ln u = \log_e u$. In particular, $\ln 1 = 0$, $\ln e = 1$, $\ln(e^x) = x$, and $e^{\ln u} = u$ for $u > 0$. The reason this base is especially useful will appear in differentiation.

4. Logarithm laws and domain checks

Let $a > 0$, $a \neq 1$, and $u, v > 0$. If $r = \log_a u$ and $s = \log_a v$, then

$$uv = a^r a^s = a^{r+s}, \quad \log_a(uv) = r + s = \log_a u + \log_a v.$$

The corresponding quotient and power arguments give

Logarithm laws | $u, v > 0, t \in \mathbb{R}$

$$\begin{aligned} \log_a(uv) &= \log_a u + \log_a v, & \log_a(u/v) &= \log_a u - \log_a v, \\ \log_a(u^t) &= t \log_a u. \end{aligned}$$

To change base, write $a^x = u$ and take natural logarithms:

$$x \ln a = \ln u, \quad \boxed{\log_a u = \frac{\ln u}{\ln a}}.$$

The denominator is nonzero. Also $a^x = e^{x \ln a}$ for $a > 0$.

Example

Let $h(x) = 3 + 2^{x-1}$, $x \in \mathbb{R}$. It is strictly increasing with range $(3, \infty)$. From $y - 3 = 2^{x-1}$,

$$h^{-1}(y) = 1 + \log_2(y - 3), \quad y > 3.$$

Its range is $\mathbb{R}$. The argument restriction records the original range.

Example

Solve $\ln(x-1) + \ln(x+1) = \ln 8$. First require $x > 1$. On that domain, the product law gives

$$\ln(x^2 - 1) = \ln 8 \implies x^2 - 1 = 8.$$

The algebraic candidates are ± 3 , but only $x = 3$ lies in the original domain. At $x = -3$, the original logarithms are not defined.

Two identities to reject. There is no sum rule: $\ln(1+1) = \ln 2 \neq 0 = \ln 1 + \ln 1$. Also, $\ln(x^2) = 2 \ln |x|$ for $x \neq 0$, whereas $2 \ln x$ is defined only for $x > 0$.

Example

Taking natural logarithms in $e^{2x-1} = 5$ gives $2x - 1 = \ln 5$, hence $x = (1 + \ln 5)/2$. For $\log_{1/2} x > 2$, first require $x > 0$. Since $2 = \log_{1/2}(1/4)$ and this logarithm is decreasing, the solution is $0 < x < 1/4$.

5. Inverse trigonometric functions

Sine, cosine, tangent, and secant repeat their values, so each needs a restricted domain before it can be inverted. The conventional choices are below. All angles are in radians.

Inverse	Domain	Range of the inverse
$\arcsin$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\arccos$	$[-1, 1]$	$[0, \pi]$
$\arctan$	$\mathbb{R}$	$(-\pi/2, \pi/2)$
arcsec	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] \setminus \{\pi/2\}$

For example,

$$\arcsin u = \theta \iff \sin \theta = u \text{ and } -\pi/2 \leq \theta \leq \pi/2.$$

This definition chooses a unique angle from the specified interval. The analogous conditions define $\arccos$ and $\arctan$. For $\sec \theta = 1/\cos \theta$, we use the restriction $\theta \in [0, \pi] \setminus \{\pi/2\}$. Its inverse is arcsec ; equivalently, $\operatorname{arcsec} u = \arccos(1/u)$ for $|u| \geq 1$.

The two cancellation directions have different domains

$$\begin{aligned}\sin(\arcsin u) &= u \quad (-1 \leq u \leq 1), \\ \arcsin(\sin \theta) &= \theta \quad (-\pi/2 \leq \theta \leq \pi/2).\end{aligned}$$

Outside that interval the second formula is not valid. Always check the range selected for the inverse.

Example

Since $\sin(5\pi/6) = 1/2$,

$$\arcsin(\sin(5\pi/6)) = \arcsin(1/2) = \pi/6.$$

The inverse returns $\pi/6$, the angle in its range, rather than $5\pi/6$. Likewise, $\arccos(\cos(4\pi/3)) = 2\pi/3$.

Example

Let $\theta = \arctan x$. Then $\tan \theta = x$ and $-\pi/2 < \theta < \pi/2$, so $\cos \theta > 0$. The identity $1 + \tan^2 \theta = 1/\cos^2 \theta$ gives

$$\cos(\arctan x) = \frac{1}{\sqrt{1+x^2}} \quad (x \in \mathbb{R}).$$

The positive sign comes from the range restriction, including when $x < 0$.

Two different selected angles. The chosen ranges give

$$\arcsin(\sin(3\pi/4)) = \pi/4, \quad \arctan(\tan(3\pi/4)) = -\pi/4.$$

The first answer belongs to $[-\pi/2, \pi/2]$; the second belongs to $(-\pi/2, \pi/2)$.

Reading. Stewart, *Calculus: Early Transcendentals*, 9th ed., Section 1.3 for operations and composition, and Section 1.5, pp. 54–63, for inverse functions, logarithms, and inverse trigonometric functions.